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Strip Theory for Setting Fin Angles (rev. 3)

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Introduction

This problem began with the earliest fin stabilized sounding rockets. Fin warpage causes such rockets to respond, in general, by randomly rolling subject to random pitch/yaw moments. This note documents a process to bring the roll rate and the pitch/yaw fin misalignment moments under control. The motivation for this is twofold:

- To minimize trajectory dispersions (departures from nominal) caused by fin misalignments. Even more importantly, a well-chosen roll rate profile will tend to "average out" the impact point errors arising from all kinds of body-fixed misalignments, especially thrust misalignment.
- To minimize airframe structural loading, especially on the fins themselves. But, fin misalignments contribute to the angle of attack, and hence, to loading to all airframe components.

The fin setting process attempts to adjust fin cant angles to achieve the desired roll rate while minimizing the perturbing pitch/yaw moments. The first step in setting is to measure the pre-existing fin cant angles. Then, using the results of this note, modify the cant angles to meet the stated objectives. When setting the fin cant angles, it will be shown that all fin panels should have the same angle, and that common angle should cause the planned roll rate to occur.

Rolling Moment

First, consider strip theory: Each fin panel is decomposed into a set of streamwise strips, each of which has the same normal force coefficient slope. No fin-body interference is modeled. No variation in local normal force coefficient slope due to tip losses is modeled. This is the simplest possible aerodynamics model accounting only for fin planform geometry.

Consider the fin planform as sketched in Figure 1: Then , define the notation to be used:

Mnemonic	Definition
N_T	Fin panel normal force, lb,
N	Number of fin panels,
q	Dynamic pressure, lb/ft^2,
U	Free Stream velocity, ft/sec.
h	Altitude, ft AGL,
L	Launcher length, ft,
Mnemonic	Definition

I_{xx} Roll moment of inertia, sl-ft^2,
ρ Atmospheric mass density, sl/ft^3,
M Free stream Mach number,
α Local (strip) angle of attack, rad,
p Roll rate, rad/sec,
R Body radius, ft,
b Fin semispan, ft,
c Local fin chord, ft,
c_R Root chord, ft,
c_T Tip chord, ft
S Aerodynamic reference area, ft^2 $=\pi R^2$,
d Aerodynamic reference length, ft, $=2R$,
$c_{N\alpha}$ Fin airfoil normal force coefficient slope, per rad,
δ Fin cant angle, rad,
L Rolling moment, ft-lb, and
y Spanwise coordinate, ft.

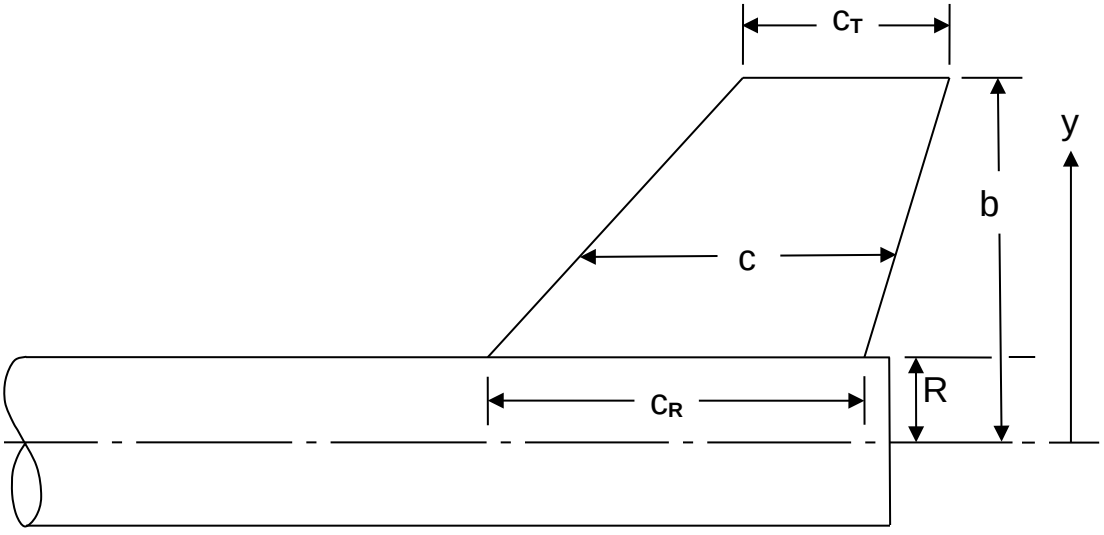


Figure 1

Now, consider a chordwise strip of span dy. The normal force dN_T acting on the strip (positive as sketched in Fig. 2) is

$dN_T = qcdyc_{N\alpha}(\delta - py/U)$, and

$dL = ydN_T = qycdyc_{N\alpha}(\delta - py/U)$

Note here that according to the thin airfoil theories, $c_{N\alpha}$ depends only on the Mach number: But, a better way to view this is to take an average over the entire panel, thus capturing the effects of root and tip losses.

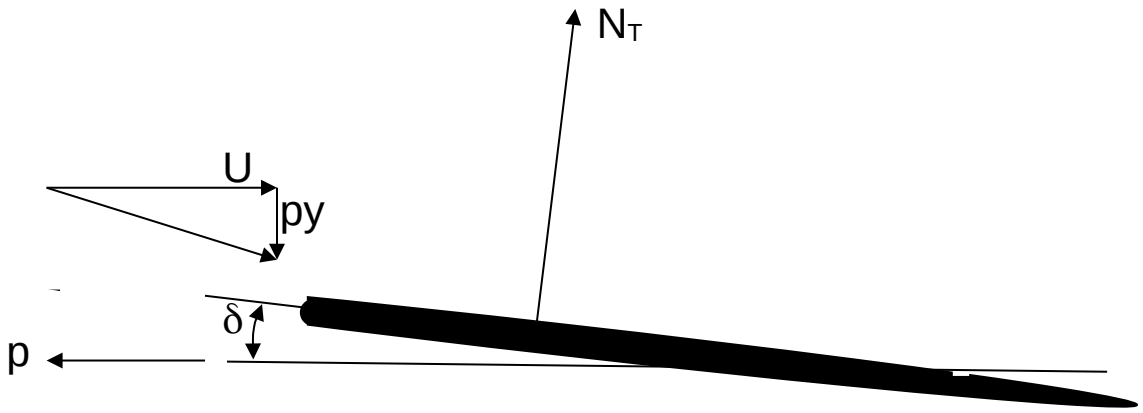


Figure 2

Next, assume the spanwise variation of the chord is linear:

$$c = A - Cy \text{, where}$$

$$A = \frac{c_R b - c_T R}{b - R} \text{, and}$$

$$C = \frac{c_R - c_T}{b - R}$$

With this result, the expression for the normal force and rolling moment on a single fin panel can be found from integration to be

$$N_T = qc_{N\alpha}(b - R)\delta\left[A - C(b+R)/2\right] + qc_{N\alpha}(b - r)(p/U)\left[- A(b+r)/2 + C(b^2 + bR + R^2)/3\right]$$

,
and the rolling moment is

$$L = qc_{N\alpha}(b - R)\delta\left[A(b+R)/2 - C(b^2 + bR + R^2)/3\right]$$
$$+ qc_{N\alpha}(b - R)(p/U)\left[- A(b^2 + bR + R^2)/3 + C(b^3 + b^2R + bR^2 + R^3)/4\right].$$

The spanwise location of the center of pressure Y_{CP} due to δ can easily be found to be

$$Y_{CP} = \frac{3A(b+R) - 2C(b^2 + bR + R^2)}{6A - 3C(b+R)}.$$

The classical rolling stability derivatives can be based on body cross section area and diameter;

$C_L = \frac{L}{qSd} = \frac{L}{2\pi R^3 q}$. Then,

$$C_{L\delta} = \frac{\partial C_L}{\partial \delta} = \frac{c_{N\alpha}(b - R)}{2\pi R^3}\left[A(b + R)/2 - C(b^2 + bR + R^2)/3\right].$$

The roll damping derivative includes the torques from all **N** fin panels:

$$C_{Lp} = \frac{\partial C_L}{\partial (pd/2U)} = \frac{\partial C_L}{\partial (pR/U)}, \text{ or}$$

$$C_{Lp} = \frac{c_{N\alpha}(b - R)N}{2\pi R^4}\left[- A(b^2 + bR + R^2)/3 + C(b^3 + b^2 R + bR^2 + R^3)/4\right]$$

Fin Adjustment

The objective driving the fin adjustment process is to modify the pre-existing fin cant angles to reduce their combined pitch and yaw moments to zero and to achieve the desired roll rate. The pitch and yaw moments from fin cant errors contribute to both trajectory dispersion (departure from the planned trajectory) and structural loading. No good ever comes from them. Obtaining the desired roll rate profile is essential to minimizing trajectory dispersions, a very good thing indeed.

While there are many possible fin adjustment solutions, adjusting the cant angles of all fin panels to a common value after nulling the panel normal force is the preferred approach. This can be easily shown for three, or more, panels to reduce the combined pitch & yaw moments to zero, and it will result in all panels having the same cant angle-induced aerodynamic loads.

Now, suppose we level the rocket and measure the “as built” cant angle on a fin. See Fig. 3 below for a sketch of this process. Start by placing the rocket in its cradle and leveling its body with a carpenter's laser level. Roll the body until the first fin panel is level as found when the laser level is oriented in the spanwise direction. Next rotate the laser level until it points along the body axis. Since the fin will have some initial cant angle we expect the laser spot observed on a screen at the body nose to be higher or lower than the nose tip. Adjust the fin cant angle until the laser spot shows the desired height relative to the nose tip. Repeat for all the other fin panels. These results are captured in CANT ANGLE2.xls.

A typical example is described by calculating the fin cant angle from strip theory so that the burnout roll rate = 0.3 * the burnout pitch natural frequency. This gives about 5 roll cycles from liftoff to burnout. The nominal cant angle $\delta = 0.2^\circ$ meets the ~0.9 hertz condition.

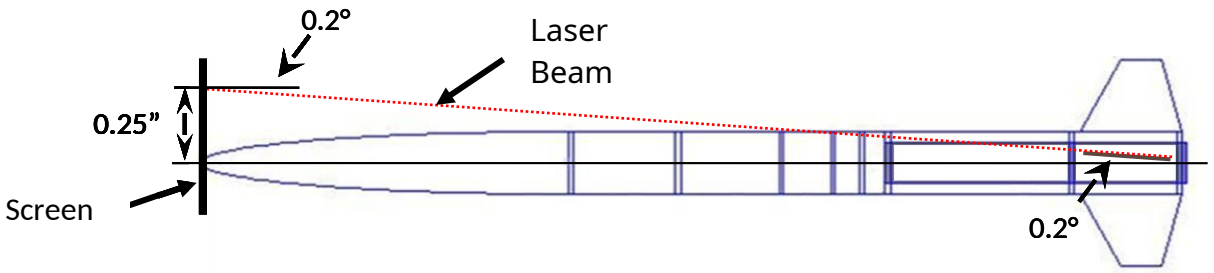


Figure 3

The fin cant angle[1] needed to produce a roll rate p is

$$\delta = \frac{-pRC_{Lp}}{NUC_{L\delta}}(1-J),$$

where

$$J = \frac{I_{xx}}{2\pi\rho(h-L)R^4 C_{Lp}}.$$

For small rockets up to about, say 8" diameter, this is as about as good as it's going to be.

For larger rockets, curve fitting the cant angle data might be a better way to go. The simplest curve to fit to two measurements at two different spanwise stations is a simple helix:

$$\delta(y) = \frac{-\delta_1 y_2 + \delta_2 y_1}{y_1 - y_2} + \frac{\delta_1 - \delta_2}{y_1 - y_2} y.$$

The normal force and rolling moment in the first term is just that found with δ replaced with $\frac{-\delta_1 y_2 + \delta_2 y_1}{y_1 - y_2}$. Similarly, if p/U is replaced by $\frac{\delta_1 - \delta_2}{y_1 - y_2}$, the result is the normal force and rolling moment due to the second term. As before, we adjust by nulling out each panel's normal force, and then add in δ needed to achieve the desired roll rate.

Roll Rate Selection

For multistage rockets, first establish the last stage fin cant angle, then the penultimate stage fin cant angle, and so on. There are three possible choices for selecting the booster roll rate and fin cant angle. Only two of these apply to upper stage(s). For a single stage rocket the mean fin cant angle $\bar{\delta}_F$ for a given λ_R is:

[1] Hoult, C. P., "Small sounding Rocket Boost Phase Roll Rate" RS&T memo, 28 September 2013

$$\lambda_R = - \frac{C_{L\delta} N \bar{\delta}_F}{R C_{Lp}} \left[\frac{2\pi\rho(h - L) R^4 C_{Lp}}{2\pi\rho(h - L) R^4 C_{Lp} - I_{xx}} \right] \quad .$$

This equation also applies to a two stage configuration if

$C_{Lp} = \sum C_{Lp}$ and $C_{L\delta} \bar{\delta}_F = \sum C_{L\delta} \delta_F$ where the summations extend over all stages.

The first choice for the booster $\bar{\delta}_F$ is to set the total number of roll revolutions to an integer value to minimize trajectory dispersions arising from body-fixed perturbations. Trajectory dispersion considerations are relatively far less important for the upper stages, and, so this does not greatly apply to them. Then, the ideal the boost phase roll angle at booster burnout would be an integer number M of roll cycles. That is,

$$\Delta\phi_{BO} = 2\pi M = - \frac{C_{L\delta} N \bar{\delta}_F}{R C_{LP}} \left[h - L + k \log\left(\frac{h - L - k}{- k} \right) \right],$$

where

$$k = \frac{I_{XX}}{2\pi\rho R^4 C_{Lp}} .$$

Again, this result also applies to a two stage rocket if the codicils developed for the above equations above apply. M should be at least 2 or 3. More is better. Once M has been selected, the above results can be used to establish the mean fin cant angle $\bar{\delta}_F$.

The second possible choice, applicable to any stage, is to select λ_R to be an inputted fraction of the pitch wavenumber, λ_P. There are two conflicting considerations. First, the body-fixed perturbations elicit an angle of attack (and structural loads) response proportional to $\frac{\lambda_P^2}{\lambda_P^2 - \lambda_R^2}$. Thus, roll rate tends to "destroy" the pitch static stability. However, as long as λ_R is less than 30-40% of λ_P the damage is not too great. Second, as noted above, a higher roll wavenumber will reduce trajectory dispersions.